CH365 Chemical Engineering Thermodynamics

Lesson 30

Generalized Property Correlations

WPR3 WARNO

- WPR3 is Wednesday 18 November

- 3 problems, 200 points, from Chapter 5 and 6
    - Calculation of ideal enthalpy and entropy changes
    - Calculation of residual enthalpy and entropy
    - Combining formation, ideal and residual properties to get real properties
    - Lost work (as in heat exchangers)
    - Derivations – property relations, Maxwell relations, generating functions
- 55 minutes, during class hour.
- Open note, open book, open computer. Calculator, Mathematica authorized.
- Upload all exam e-work to SharePoint.

Today's Agenda

- CDP IPR guidance
- Continue discussion of residual properties
- Gneralized methods
- Mixtures

CDP – IPR

- Friday 20 November by 2359, pass/fail. Failing grades will remain until repaired.
- Keep it short and simple. Answer Questions 1-2. Show progress toward one other problem.
- Signed cover sheet, PDF of Mathematica combined into single signed PDF (with signed cover sheet).
- Question 1 – Txy plot in CHEMCAD using NRTL. Suggest you place a screen shot of CHEMCAD results plot into Mathematica before bundling pdf.
- Question 2 – Txy plot in MMA for ideal case. Suggestion: Txy plot in CHEMCAD using ideal, with screenshot into Mathematica, to show what "right looks like."
- Ideal Txy plot was covered in CH362 and CH363. Problem 13.1 in CH365 is a guide.

<u>Bubble Point:</u>

x_i is known

$$y_i = K_i \cdot x_i$$

$$\sum_i y_i = 1$$

<u>Dew Point:</u>

y_i is known

$$x_i = y_i \div K_i$$

$$\sum_i x_i = 1$$

Antoine Eq.
CC Eq. 107

$$K_i = f(T,P) = \frac{P_i^{sat}}{P}$$

Bubble Point Calculation

- At the bubble point an infinitesimal amount of liquid evaporates.
- Let V be the amount of vapor phase and F the total amount of fluid.
- Let x be the liquid mole fraction and z the total (bulk) mole fraction.
- At the bubble point, almost all of the fluid is in the liquid phase.
- This means V/F is very close to zero and the liquid-phase mole fraction x is almost z.
- So, we let the bubble point V/F equal 0 and the liquid-phase mole fraction x equal z.
- That is, at V/F=0, x=z, and $\Sigma x=1$.
- Meanwhile, in the vapor phase, y=Kx and $\Sigma y=1$.
- V/F is sometimes referred to as Ψ, so $\Psi=V/F=0$ at the bubble point.
- So at the bubble point, the conditions are: $\Psi=V/F=0$, x=z, y=Kx, and $\Sigma y=1$.
- Since K is a function of temperature, we solve for the temperature that satisfies this condition.

Txy Code


```
(*VAPOR PRESSURE AND K VALUES*)
(*Eq107 and constants from CHEMCAD for calculating K*)
vp1 = {a → 88.134, b → -8498.6, c → -9.0766, d → 8.3303 * 10^-18, e → 6};
vp2 = {a → 72.55, b → -7206.7, c → -7.1385, d → 4.046 * 10^-6, e → 2};
vp = {vp1, vp2};
Psat = .001 * Exp[a + b/(t + 273.15) + c * Log[t + 273.15] + d * (t + 273.15)^e];
Ptot = 101.325 ; (*bar*)
K = (Psat /. vp) / Ptot;

(*BUBBLE POINT CALCULATION*)
(*ξ is the specified feed mole fraction so z=ξ and x=z=ξ*)
bx1 = ξ;
bx2 = 1 - ξ;

(*equilibrium expressions to give by1 and by2 from bx1 and bx2*)
(*system of 3 eqns and 3 unks for FindRoot*)
bf1 = bx1 * K[[1]] - by1 == 0; (*by1 unknown*)
bf2 = bx2 * K[[2]] - by2 == 0; (*by2 unknown*)
bf3 = by1 + by2 == 1; (*sum of mole fractions in vapor*)
bfs = {bf1, bf2, bf3};

(*initial values for Findroot*)
t0 = 100.; (*initial guess for t for FindRoot*)
by10 = .5; (*initial guess for by1 for FindRoot*)
by20 = .5; (*initial guess for by2 for FindRoot*)
(*solve system of equations*)
Tb[arg_] := t /. FindRoot[bfs /. ξ → arg, {{t, t0}, {by1, by10}, {by2, by20}}];
```

Dew Point Calculation

- At the dew point an infinitesimal amount of vapor condenses.
- Let V be the amount of vapor phase and F the total amount of fluid.
- Let y be the vapor mole fraction and z the total (bulk) mole fraction.
- At the dew point, almost all of the fluid is in the vapor phase.
- This means V/F is very close to one, and the vapor-phase mole fraction y is almost z.
- So, we let the dew point V/F equal 1 and the vapor-phase mole fraction y equal z.
- That is, at V/F=1 (Ψ=1), y=z and Σy=1.
- Meanwhile, in the liquid phase, x=y/K and Σx=1.
- So, at the dew point, the conditions are: Ψ=V/F=1, y=z, x=K/y, and Σx=1.
- Since K is a function of temperature, we solve for the temperature that satisfies this condition.

Residual Properties

superscript “R” refers to “residual”

$$M^R \equiv M - M^{ig} \qquad \text{(Eq. 6.41)}$$

- M = V, U, H, S, or G
- M and M^{ig} are at the same T and P

$$H^R = H - H^{ig} \qquad S^R = S - S^{ig} \qquad G^R = G - G^{ig}$$

$$H = H^{ig} + H^R \qquad S = S^{ig} + S^R \qquad G = G^{ig} + G^R$$

$$dH^{ig} = C_P^{ig}\,dT \qquad \text{(Eq. 6.23)}$$

$$dS^{ig} = \frac{C_P^{ig}}{T}dT - R\cdot\frac{1}{P}\cdot dP \qquad \text{(Eq. 6.24)}$$

$$H^{ig} = H_0^{ig} + \int_{T_0}^{T} C_P^{ig}\,dT \qquad S^{ig} = S_0^{ig} + \int_{T_0}^{T} \frac{C_P^{ig}}{T}dT - R\cdot\ln\left(\frac{P}{P_0}\right)$$

$$V^R = V - V^{ig} \qquad V = \frac{ZRT}{P} \qquad V^{ig} = \frac{RT}{P}$$

$$V^R = \frac{ZRT}{P} - \frac{RT}{P} = \frac{RT}{P}(Z-1) \qquad \text{(Eq. 6.40)}$$

- The true worth of the equations for ideal gases is now evident.
- They are important because they provide a convenient base for the calculation of real gas properties.

Residual Properties from Cubic EOS

Generic cubic equation of state (in Z-Form) **(WPR3)**

$$q = q(x) = \frac{\Psi\alpha(x)}{\Omega \cdot x} \qquad \text{(Eq. 3.51)}$$

$$\alpha = \alpha(x)$$

(α is found in Table 3.1)

These are the red equations from L28 slide 8 (cleanup).

Important: α is written as a function of x where x replaces T_r.

q derivatives must be evaluated before process conditions (T_R, P_r) are entered.

Problem 6.141 in PS10

$$\beta = \Omega\frac{P_r}{T_r} \qquad (3.50)$$

$$Z = 1 + \beta - q\beta\frac{Z-\beta}{(Z+\varepsilon\beta)(Z+\sigma\beta)} \qquad (3.48)$$

$$\frac{H^R}{RT} = Z - 1 + T_r\left(\frac{dq}{dT_r}\right)\cdot I$$

(page 497)

$$\frac{S^R}{R} = \ln(Z-\beta) + \left(q + T_r\frac{dq}{dT_r}\right)\cdot I$$

$$I = \frac{1}{\sigma-\varepsilon}\ln\left(\frac{Z+\sigma\beta}{Z+\varepsilon\beta}\right) \qquad (13.72)$$

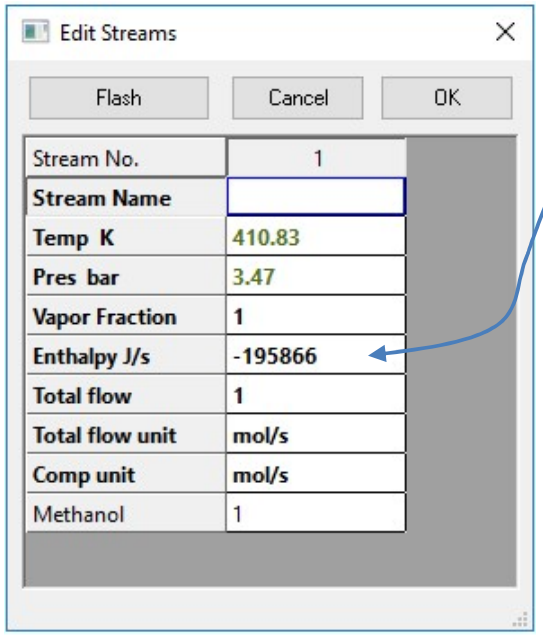
$$\varepsilon \neq \sigma$$

Connection to Total Properties

Add ideal gas and residual

$$H = H_0^{ig} + R\int_{T_0}^{T} C_P^{ig} dT + H^R \quad \textbf{(Eq. 6.50)}$$

CHEMCAD



$$H_1 = H_0^{ig} + R\int_{T_0}^{T_1} C_P^{ig} dT + H_1^R \qquad H_2 = H_0^{ig} + R\int_{T_0}^{T_2} C_P^{ig} dT + H_2^R$$

$H_0^{ig} = \Delta H_{f298}^{o}$ = standard formation enthalpy, App C, pp 671-672

Difference between streams:

residual property at T_2

residual property at T_1

$$\Delta H = H_2 - H_1 = R\int_{T_1}^{T_2} C_P^{ig} dT + H_2^R - H_1^R \quad \text{(Eq. 6.72)}$$

For CHEMCAD entropy, use Report -> Stream Reports -> Customize Property Sets -> Overall

$$S = S_0^{ig} + \int_{T_0}^{T} C_P^{ig}\frac{dT}{T} - R\ln\frac{P}{P_0} + S^R \quad \textbf{(Eq. 6.51)} \qquad \Delta S = \int_{T_1}^{T_2} C_P^{ig}\frac{dT}{T} - R\ln\frac{P_2}{P_1} + S_2^R - S_1^R \quad \text{(Eq. 6.73)}$$

$S_0^{ig} = \Delta S_{f298}^{o}$, also from App C, pp 660-662, from ΔH_{f298}^{o} and ΔG_{f298}^{o}

CHEMCAD standard enthalpy and Gibbs energy are in Component Data -> Basic Data

$$\Delta S_{f298}^{ig} = \frac{\Delta H_{f298}^{ig} - \Delta G_{f298}^{ig}}{298.15}$$

Three Methods for Residual Properties

Cubic Equations of States (L28 & L29):

- RK, SRK, PR EOS from Table 3.1

Generalized Methods (New for Today):

- Lee-Kesler Tables
- Virial Equation of State

Lee-Kesler Table Method

Valid for liquids, vapors and gases

$$Z = Z^0 + \omega Z^1$$ (Pitzer correlation, Eq. 3.53)

$$\frac{H^R}{RT_c} = \frac{\left(H^R\right)^0}{RT_c} + \omega \frac{\left(H^R\right)^1}{RT_c}$$ (Eq.6.66)

$$\frac{S^R}{R} = \frac{\left(S^R\right)^0}{R} + \omega \frac{\left(S^R\right)^1}{R}$$ (Eq. 6.67)

Z^0, Z^1, $\frac{\left(H^R\right)^0}{RT_c}$, $\frac{\left(H^R\right)^1}{RT_c}$, $\frac{\left(S^R\right)^0}{R}$, $\frac{\left(S^R\right)^1}{R}$, ϕ^0, and ϕ^1

Appendix D
pages 676-692

Virial Equation

Truncated

$$Z = 1 + \frac{BP}{RT} = 1 + \hat{B}\frac{P_r}{T_r} \qquad \text{(Eq.3.57)}$$

$$\hat{B} = \frac{BP_c}{RT_c} \qquad \text{(Eq.3.58)}$$

Second virial coefficient

$$\hat{B} = B^0 + \omega B^1 \qquad \text{(Pitzer Correlation, Eq. 3.59)}$$

$$\frac{H^R}{RT_c} = P_r\left[B^0 - T_r\frac{dB^0}{dT_r} + \omega\left(B^1 - T_r\frac{dB^1}{dT_r}\right)\right] \qquad \text{(Eq. 6.68)}$$

$$\frac{S^R}{R} = -P_r\left(\frac{dB^0}{dT_r} + \omega\frac{dB^1}{dT_r}\right) \qquad \text{(Eq. 6.69)}$$

$$B^0 = 0.083 - \frac{0.422}{T_r^{1.6}} \qquad \text{(Eq. 3.61)}$$

$$B^1 = 0.139 - \frac{0.172}{T_r^{4.2}} \qquad \text{(Eq. 3.62)}$$

$$\frac{dB^0}{dT_r} = \frac{0.675}{T_r^{2.6}} \qquad \text{(Eq. 6.70)}$$

$$\frac{dB^1}{dT_r} = \frac{0.722}{T_r^{5.2}} \qquad \text{(Eq. 6.71)}$$

Mixtures of Gases I - Textbook

The critical temperature of a gas is the maximum temperature at which compression can cause liquefaction. The critical pressure of a gas is the minimum pressure at which cooling causes liquefaction. Mixing of gases results in new critical properties for the mixture, and the mixture behaves as a "pseudocritical" pure component.

"pseudocritical" properties

$$\omega \equiv \sum_i y_i \omega_i \qquad \text{(Eq. 6.78)}$$

"pseudocritical acentric factor"

$$T_{pc} \equiv \sum_i y_i T_{ci} \qquad \text{(Eq. 6.79)}$$

"pseudocritical temperature"

subscript "p"

$$P_{pc} \equiv \sum_i y_i P_{ci} \qquad \text{(Eq. 6.80)}$$

"pseudocritical pressure"

"pseudoreduced temperature"

$$T_{pr} = \frac{T}{T_{pc}} \qquad \text{(Eq. 6.81)}$$

"pseudoreduced pressure"

$$P_{pr} = \frac{P}{P_{pc}} \qquad \text{(Eq. 6.82)}$$

Mixtures of Gases II – API - Best

API Technical Data Book, T.T. Daubert, ed., 5th Edition, American Petroleum Institute, 1992. Standard method now widely used.

worked example is in reference, p. 4-25

"pseudocritical" properties

"true critical temperature" of mixture

$$T_{cm} = \sum_{i=1}^{n} (\theta_i \cdot T_{ci}) \qquad \text{(Eq. 4B1.1-1)}$$

volume fraction of component i

$$\theta_i = \frac{x_i \cdot V_{ci}}{V_{pc}} \qquad \text{(Eq. 4B1.1-2)}$$

molar average critical volume

$$V_{pc} = \sum_{i} x_i \cdot V_{ci} \qquad \text{(Eq. 4B1.1-3.79)}$$

"true critical pressure" of mixture

worked example is in reference, pp. 4-28 to 4-29

$$P_{cm} = P_{pc} + P_{pc}\left[5.808 + 4.93 \cdot \sum_{i=1}^{n} (x_i \cdot \omega_i)\right] \cdot \left[\frac{T_{cm} - T_{pc}}{T_{pc}}\right] \qquad \text{(Eq. 4B2.1-1)}$$

pseudocritical temperature of mixture

$$T_{pc} = \sum_{i} (x_i \cdot T_{ci}) \qquad \text{(Eq. 4B2.1-2)}$$

pseudocritical pressure of mixture

$$P_{pc} = \sum_{i=1}^{n} (x_i \cdot P_{ci}) \qquad \text{(Eq. 4B2.1-3)}$$

Questions?